

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			1. Oxygen crosses the membrane by (simple) diffusion (1) NOT facilitated 2. oxygen passes through the phospholipid (bilayer) (1) Membrane = neutral 3. Sodium ions pass through {channel/carrier} proteins (1) intrinsic proteins = neutral 4. by {facilitated diffusion/active transport/ cotransport} (1) Reject active transport in relation to channel proteins If oxygen and sodium are not treated separately, allow one mark max for correct linkage of diffusion with phospholipid bilayer or {facilitated diffusion/ active transport/ cotransport} with {channel/ carrier} proteins	4			4		
	(b)			1. The percentage ion composition of {fresh water/ habitat} is <u>lower</u> than that in the carp's blood plasma/ fresh water is hypotonic to the blood/ ORA (1) 2. The water potential of fresh water is {higher/less negative} than the water potential of the carp's blood/ ORA (1) 3. The carp takes in water (through the gills) by osmosis (1) 4. (Producing large volumes of dilute urine) prevents {osmotic lysis/ bursting} of cells (1) 5. Maintaining ion composition (1) Accept reduces the loss of ions		1 1	1 1	4		